Getting Personal with Wiivv

Wiivv Wearables, Inc./Photo Credit: Shamil Hargovan

Cofounders

SHAMIL HARGOVAN

BA, Philosophy, Politics, and Economics, Claremont McKenna College, 2010

LOUIS-VICTOR (LV) JADAVJI

BA, International Relations and Leadership Studies, Claremont McKenna College, 2015

Dialogue *with* Shamil Hargovan

FAVORITE BAND ON MY SMARTPHONE APP
Daft Punk

BEST ADVICE I'VE RECEIVED
Everything in life should have a reason

MY BIGGEST SURPRISE AS AN ENTREPRENEUR
How difficult having a personal life would be

MY ADVICE FOR NEW ENTREPRENEURS
You're not a true entrepreneur until you figure out how to make everyone around you an entrepreneur for your vision

FIRST ENTREPRENEURIAL EXPERIENCE
Turned college internship into start-up idea (Kicklight.com) at Stanford Research Institute

MY FAVORITE SMARTPHONE APP
Wunderlist

MY BIGGEST WORRY AS AN ENTREPRENEUR
Staying true to the vision in spite of short-term tactical ups and downs

BEST PART OF BEING A STUDENT
The freedom and time to figure out who you are and what you stand for

WHAT I DO WHEN I'M NOT WORKING
Enjoy the outdoors with my wife Jillian and our dog Xena

CHAPTER 2

Recognizing *Opportunities* and Generating Ideas

OPENING PROFILE

WIIVV

Solving Foot Pain and Fatigue via Custom Fitted 3D Printed Insoles

• *Web: https://wiivv.com* • *Facebook: wiivvit* • *Twitter: @wiivvit*

MyLab Entrepreneurship

Improve Your Grade!

If your instructor is using MyLab Entrepreneurship, visit **www.pearson.com/mylab/entrepreneurship** for videos, simulations, and writing exercises.

Shamil Hargovan and Louis-Victor (LV) Jadavji met at a 3D printing conference in 2013. They quickly discovered that they had two things in common: first, they were both affiliated with Claremont McKenna College in California—Hargovan as an alumni and Jadavji as a student—and second, they both had an active interest in 3D printing.

After the conference, Hargovan and Jadavji kept in touch. At the time, Hargovan worked for Hewlett-Packard as a Group Product Manager and 3D Printing Lead. Jadavji remained in school and worked for a software company in the San Francisco Bay area. The two continued to share their interest in 3D printing. Hargovan was kicking around several ideas, outside his job at HP, including using 3D printing to create products for athletes, such as helmets. In early 2014, Jadavji had a bad experience with a pair of orthotics, and the thought occurred to him to utilize 3D printing to create customized, affordable insoles that would provide users maximum support.

In April of that year, Hargovan and Jadavji decided to work together on the 3D printed insoles idea. The idea made sense on several levels. First, many people have trouble with their feet and struggle to find a solution. Individuals who spend hours a day standing or walking, such as nurses and security guards, are the most susceptible. Second, there was a gap in the marketplace for products to help people with foot fatigue and other issues. On one end of the market were custom orthotics, which cost $400 or more. Along with the cost, being fitted for orthotics usually requires a visit to a podiatrist and is a time-consuming process. On the other end of the market were inexpensive insoles, which are basically small cushions you put in your shoes. Because the insoles aren't custom fitted, they often don't fit right or provide adequate support. What was missing was a product in the middle—a quality, custom fitted insole that could be obtained quickly at an affordable price. That was the sweet spot that Hargovan and Jadavji felt 3D printing was ideally suited to accommodate.

Wiivv, the company the two started, took shape in 2014 and 2015. It took time for the idea to be fleshed out and for the manufacturing process to be developed and the product to be tested. To test the process and product, Wiivv assembled a group of people called the Wiivv 100. These were people who spent a lot of time on their feet, and who beta tested the custom printed insoles Wiivv was 3D printing. The group was utilized for

LEARNING OBJECTIVES

After studying this chapter you should be ready to:

1. Explain the difference between opportunities and ideas.
2. Describe the three general approaches entrepreneurs use to identify opportunities.
3. Discuss the personal characteristics of entrepreneurs that contribute to their ability to recognize business opportunities.
4. Identify and describe techniques entrepreneurs use to generate ideas.

six months and saw the product through 72 iterations. Some of the testers were surveyed and others were brought into Wiivv's offices to be interviewed and for their insoles to be examined (after various lengths and types of use). The Wiivv 100 ultimately became strong advocates for the product and spread the word about Wiivv via social media.

Wiivv launched in early 2016 via a Kickstarter campaign. The campaign raised $235,054 and had 2,765 backers. The idea for the campaign wasn't so much to raise money, but to validate that there was a market for high-quality, 3D-printed custom insoles at a modest price. If you'd like to look at Wiivv's Kickstarter campaign, which includes a video, go to www.kickstarter.com and type in Wiivv. The campaign was successful and was the largest Kickstarter campaign at the time for a 3D printed product. The market validation that the campaign provided also helped Wiivv raise money from investors down the road and generated substantial positive publicity.

The way Wiivv works is really cool. To purchase a Wiivv custom insole, you start by going to Wiivv's website to buy a purchase code. The code costs $69 or $89, depending on the insole you buy. You then download the Wiivv app, watch a short tutorial, and tailor the insole to your liking. You can pick from various colors (for the bottom) and patterns (for the top). You then take five photos of your feet using your smartphone. The app guides you through the process, which doesn't take more than a few minutes. You finish up via a standard checkout process. In 7–14 days, your custom insoles arrive in the mail. Using the insoles is easy. You just place them in your shoes. You can slip the insoles in and out of all the shoes you own. Most people immediately notice a difference in terms of less foot fatigue. If you are not completely satisfied, Wiivv provides a 30-day satisfaction guarantee.

Behind the scenes, Wiivv's operations are complex. To manufacture the insoles, the firm uses a process called adaptive manufacturing. The part of the insole that provides a custom fit is 3D printed, and the rest is built via traditional manufacturing processes. A lot of thought regarding the user experience went into how the insoles are ordered and the photos are taken. Once the photos reach Wiivv, they are turned into data that Wiivv's commercial-grade 3D printers can understand. All of Wiivv's products are manufactured in San Diego, CA.

To grow its business, Wiivv has raised $7.88 million through two seed rounds and a Series A round. Wiivv's ultimate vision is to add active mental years to people's lives by producing custom fit footwear, orthopedics, and wearable products that encourage them to lead a more active lifestyle. A second line of products, in addition to insoles, is in the works but has yet to be announced publicly.

In this chapter, we discuss the importance of understanding the difference between ideas and opportunities. While ideas are interesting and can intrigue us as possibilities, not every idea is in fact the source of an opportunity for an entrepreneur to pursue. In addition to describing the differences between ideas and opportunities, this chapter also discusses approaches entrepreneurs use to spot opportunities, as well as factors or conditions in the external environment that may result in opportunities. As you will see, too, certain characteristics seem to be associated with individuals who are adept at spotting viable business opportunities.

The Differences Between Opportunities and Ideas

LEARNING OBJECTIVE

1. Explain the difference between opportunities and ideas.

Essentially, entrepreneurs recognize an opportunity and turn it into a successful business.[1] An **opportunity** is a favorable set of circumstances that creates a need for a new product, service, or business. Most entrepreneurial ventures

are started in one of two ways. Some ventures are externally stimulated. In this instance, an entrepreneur decides to launch a firm, searches for and recognizes an opportunity, and then starts a business, as Jeff Bezos did when he created Amazon.com. In 1994, Bezos quit his lucrative job at a New York City investment firm and headed for Seattle with a plan to find an attractive opportunity and launch an e-commerce company. Other firms, such as Wiivv, are internally stimulated. In this instance, an entrepreneur recognizes a problem or an **opportunity gap** and creates a business to address the problem or fill the identified gap. In this instance, Wiivv's founders' seek to build " . . . a better foundation for every step you take with made-to-fit-you insoles and footwear."[2] A bionics company, Wiivv uses 3D-mapping technology to solve a problem of shoes not fitting customers as well as they want to be the case.

Regardless of which of these two ways an entrepreneur starts a new business, fresh opportunities are tough to spot. Identifying a product, service, or business opportunity that isn't merely a different version of something already available is difficult. A common mistake entrepreneurs make in the opportunity recognition process is picking a currently available product or service that they like or are passionate about and then trying to build a business around a slightly better version of it. Although this approach seems sensible, such is usually not the case. The key to opportunity recognition is to identify a product or service that people need and are willing to buy, not one that an entrepreneur wants to make and sell.

As shown in Figure 2.1, an opportunity has four essential qualities: It is (1) attractive, (2) timely, (3) durable, and (4) anchored in a product, service, or business that creates or adds value for its buyer or end-user. For an entrepreneur to capitalize on an opportunity, its **window of opportunity** must be open. The term *window of opportunity* is a metaphor describing the time period in which a firm can realistically enter a new market. Once the market for a new product is established, its window of opportunity opens. As the market grows, firms enter and try to establish a profitable position. At some point, the market matures, and the window of opportunity closes. This is the case with Internet search engines. Yahoo, the first search engine, appeared in 1995, and the market grew quickly, with the addition of Lycos, Excite, and several others. Google entered the market in 1998, sporting advanced search technology. Since then, the search engine market has matured, and the window of opportunity is less prominent. Today, it would be very difficult for a new start-up search engine firm to be successful unless it offered compelling advantages over already established competitors or targeted a niche market in an exemplary manner. Bing, Microsoft's search engine, is gaining ground with approximately 20.9 percent market share (compared to 63.9 percent for Google), but only after Microsoft has exerted an enormous amount of effort in head-to-head competition with Google.[3]

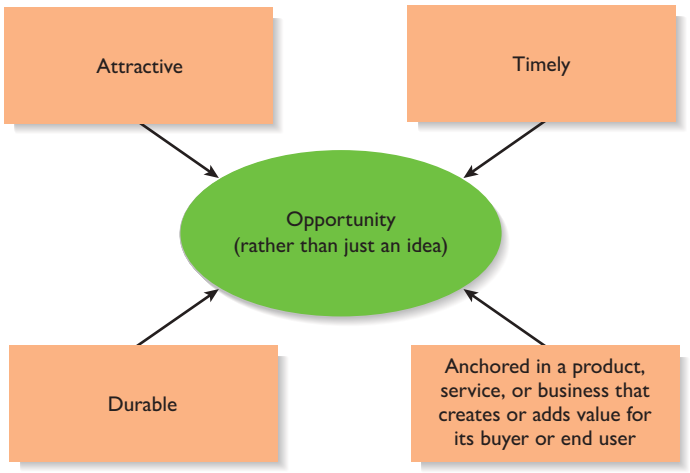


FIGURE 2.1
Four Essential Qualities of an Opportunity

It is important to understand that there is a difference between an opportunity and an idea. An **idea** is a thought, an impression, or a notion. An idea may or may not meet the criteria of an opportunity. This is a critical point because many entrepreneurial ventures fail not because the entrepreneurs that launched them didn't work hard, but rather because there was no real opportunity to begin with. Before getting excited about a business idea, it is crucial to understand if the idea fills a need and satisfies the criteria for an opportunity.

Three Ways to Identify Opportunities

LEARNING OBJECTIVE

2. Describe the three general approaches entrepreneurs use to identify opportunities.

There are three approaches entrepreneurs use to identify an opportunity their new venture can choose to pursue (see Figure 2.2). Once an entrepreneur understands the importance of each approach, she or he will be much more likely to look for opportunities and ideas that fit each profile. We discuss the three approaches in the next three sections.

Observing Trends

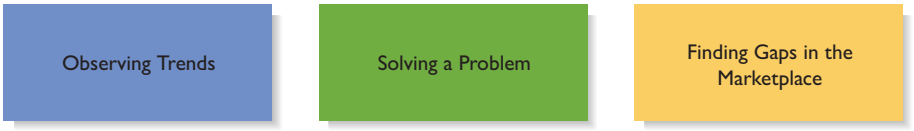
The first approach to identifying opportunities is to observe trends and study how they create opportunities for entrepreneurs to pursue. The most important trends to follow are economic trends, social trends, technological advances, and political action and regulatory changes. As an entrepreneur or potential entrepreneur, it is important to remain aware of changes in these areas. This sentiment is affirmed by Michael Yang, the founder of Become.com, a comparison shopping site, who believes that keen observation skills and a willingness to stay on top of changing environmental trends are key attributes of successful entrepreneurs:

> One of the most important attributes of a good entrepreneur is having a keen observation ability. Basically seeing what's needed in people's everyday lives and coming up with innovative new ideas and services that meet those needs . . . I always believe the entrepreneurs that anticipate trends and maintain observations of what's needed . . . to solve those needs will have a higher chance of succeeding in the marketplace.[4]

When looking at environmental trends to discern new business ideas, there are two caveats to keep in mind. First, it's important to distinguish between trends and fads. New businesses typically do not have the resources to ramp up fast enough to take advantage of a fad. Second, even though we discuss each trend individually, they are interconnected and should be considered simultaneously when brainstorming new business ideas. For example, one reason that smartphones are so popular is because they benefit from several trends converging at the same time, including an increasingly mobile population (social trend), the continual miniaturization of electronics (technological trend), and their ability to help users better manage their money via online banking and comparison shopping (economic trend). If any of these trends weren't present, smartphones wouldn't be as successful as they are and wouldn't hold as much continuing promise to be even more successful in the future.

Figure 2.3 provides a summary of the relationship between the environmental factors just mentioned and identifying opportunity gaps. Next, let's look at how entrepreneurs can study each of these factors to help them spot business, product, and service opportunity gaps.

FIGURE 2.2
Three Ways to Identify an Opportunity



StockLite/Shutterstock

As baby boomers age, opportunities will grow for firms that provide unique services to the age group. Look for the resulting expansion in organic foods, travel, and technologies that enable older people to easily connect with their families and friends.

Economic Forces. There are a number of economic forces that help determine areas that are ripe for new business ideas, as well as areas to avoid. These forces include topics such as whether consumer spending is rising or falling, whether housing starts are up or down, whether international trade is strong or weak, and whether interest rates are stable, falling, or increasing. At the time this chapter was written (spring, 2017), personal income was rising, housing starts were up, international trade was flat, and interest rates were inching up.

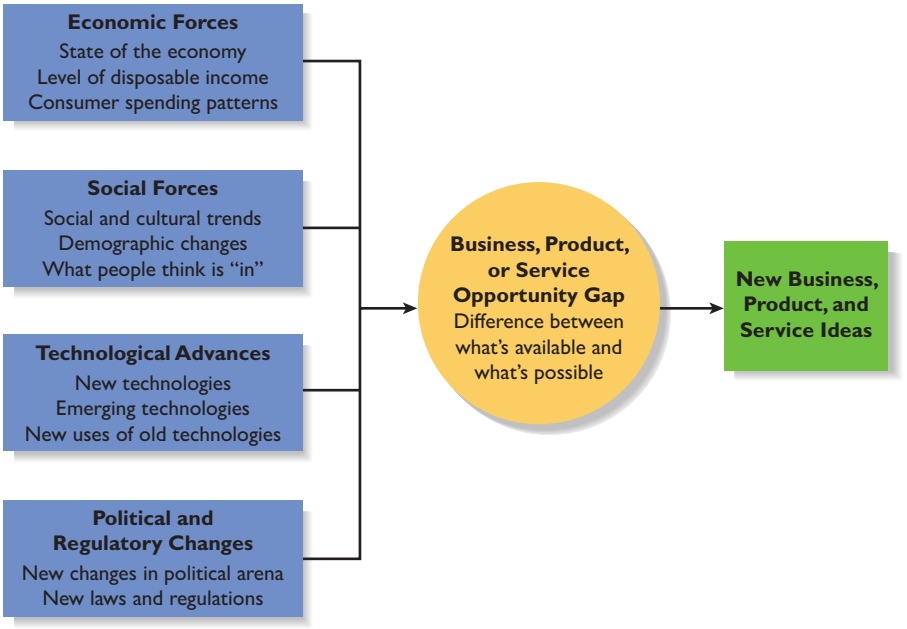


FIGURE 2.3
Environmental Trends Suggesting Business, Product, or Service Opportunity Gaps

When personal income is rising, consumers have more money to spend and are willing to buy discretionary products and services that enhance their lives. That bodes well for start-ups that provide discretionary products or services. For example, SoulCycle, the subject of Case 1.1, offers spinning classes in hip urban studios. The classes cost $34 per session. At that price, SoulCycle is more likely to gain new members when personal income is rising and the economy is strong. In contrast, when personal income is falling, not only do people have less money to spend, they are less willing to spend it, opting instead to accumulate cash. This environment provides business opportunities for start-ups that help consumers and businesses save money. An example is GasBuddy, a company started to help consumers and businesses save money on gas. Pause reading for a moment and go to GasBuddy's website, at www.gasbuddy.com, to see how the site works. A similar example of a service that helps people save money is YouNeedaBudget, which is an app that helps users reduce debt and manage their spending. Both GasBuddy and YouNeedaBudget would likely gain more users when personal incomes are falling rather than rising.

Another important economic force to evaluate is who has money to spend and what they spend it on. For example, an increase in the number of women in the workforce and their related increase in disposable income is largely responsible for the number of online retailers and boutique clothing stores targeting professional women that have opened the past several years. Similarly, the increased buying power of minority populations has resulted in an upswing of ethnic restaurants and ethnic supermarkets in the United States. Baby boomers are another group to examine. These individuals, who were born between 1946 and 1964, are retiring in large numbers and will be retiring in even larger numbers over the next five years or so. Approximately 76 million people are classified as baby boomers in the United States; currently, roughly 10,000 of them are retiring every day. The expectation is that these people will redirect a sizeable portion of their assets to products and services that facilitate their retirement. This trend will invariably spawn new businesses in many areas, largely because baby boomers have greater disposable income relative to previous generations. Areas that baby boomers spend heavily on include health care, travel, and consumer packaged goods. PillPack, the subject of "You Be the VC 2.1" in this chapter, makes a product with the ability to help baby boomers encountering difficulty when it comes to effectively dealing with their prescribed medications. The high cost of energy, coupled with a desire to be socially responsible, has also spawned a growing number of start-ups that are developing products and services that help business and consumers become more energy efficient. An example is Nest (www.nest.com), a 2010 start-up. Nest—which was acquired by Google in 2014—makes the world's first learning thermostat. The thermostat, which can be used in homes or businesses, learns from your temperature adjustments and programs itself to optimize a building's comfort and energy efficiency.[5]

An understanding of economic trends also helps identify areas to avoid. For example, a decision to launch a company that sells products or services to public schools is not always wise when economies are not growing or perhaps are declining. When tax revenues decline because economies (national, state, and/or local) are not growing, schools' budgets are cut. In turn, budget cuts reduce the ability of schools to purchase new products and services.

Social Trends. An understanding of the impact of social forces on trends and how they affect new product, service, and business ideas is a fundamental piece of the opportunity recognition puzzle. Often, the reason that a product or service exists has more to do with satisfying a social need than the more transparent need the product fills. The proliferation of fast-food restaurants, for example, isn't primarily because of people's love of fast food, but rather because people are busy and often don't have time to cook their own meals. Similarly,

social networking sites like Facebook, Snapchat, and Instagram aren't popular because they can be used to post information and photos on a website. They're popular because they allow people to connect and communicate with each other, which is a natural human tendency.

Changes in social trends alter how people and businesses behave and how they set their priorities. These changes affect how products and services are built and sold. Here is a sample of the social trends that are currently affecting how individuals behave and set their priorities.

- Aging of the population
- The increasing diversity of the population
- Millennials entering the workforce
- Growth in the use of mobile devices
- An increasing focus on health and wellness
- Emphasis on clean forms of energy, including wind, solar, biofuels, and others
- Continual migration of people from small towns and rural areas to cities
- Desire for personalization (which creates a need for products and services that people can tailor to their own tastes and needs)

Each of these trends is providing the impetus for new business ideas. The continual migration of people from small towns and rural areas to cities, for example, is creating more congestion in cities. Businesses like Zipcar, a car-sharing service, and Zagster, a bicycle-sharing service, were started in part to address this problem. Similarly, the aging of the population is creating business opportunities from vision care to home health care to senior dating sites. An example is Glaukos, a company that's developing a new approach for treating glaucoma, which is an age-related eye disorder.

The proliferation of mobile devices is a social trend that's opening business opportunities for entrepreneurs across the globe. More than 2.1 billion people worldwide owned mobile phones in 2016, and that number is expected to grow to 2.87 billion by 2020. An increasing percentage of the mobile phones are smartphones, like the Apple iPhone and Android-equipped devices.[6] The proliferation of smartphones will spawn new businesses both in the United States and throughout the world. Sometimes social trends converge to create a particuarly compelling business idea. For example, CareZone, a start-up in the health-care space, was launched by Jonathan Schwartz, who was looking for ways to better manage the care of five aging parents and in-laws. CareZone is an app for smartphones, tablets, or computers that allows those involved in a person's care to share and save information in a secure, online setting. CareZone's potential is bolstered by two societal trends: the aging of the population and the growth in the use of smartphone apps and other connected devices.

The increasing interest in social networking sites such as Facebook, Twitter, and LinkedIn is a highly visible social trend. In 2016, 78 percent of U.S. residents had a social media profile, a five percent increase from the year before. According to estimates, the number of worldwide social media users reached 1.96 billion in 2016 and is expected to grow to 2.5 billion by 2018.[7] In addition to providing people new ways to communicate and interact with each other, social networks also act as platforms for other businesses to build on. Zynga, for example, the maker of popular online games FarmVille and Scramble, became popular by making browser-based games that worked as application widgets on Facebook. Similarly, entrepreneurs have launched businesses for the purpose of starting social networks that cater to specific niches. An example is PatientsLikeMe, the subject of Case 1.2, which is a social networking site for people with serious diseases.

Technological Advances. Advances in technology frequently dovetail with economic and social changes to create opportunities. For example, there are many overlaps between an increased focus on health and wellness and technology. Wearable devices, like the Fitbit Charge 2 and the Jawbone Up, help people maintain a healthy lifestyle by monitoring their movements and sleep. In fact, according to Statista, the global market for wearable devices is expected to reach 19 billion dollars in 2018, more than 10 times its value in 2013.[8] The projected 19 billion dollar market includes wearable devices that have a variety of applications. For example, the Valedo Back Therapy is a wearable device that uses sensors to detect the nature of a user's lower back pain. Based on the data the sensors collect, the device suggests exercises the user can complete in a game format to relieve the pain.[9]

Technological advances also provide opportunities to help people perform everyday tasks in better or more convenient ways. For example, OpenTable.com is a website that allows users to make restaurant reservations online and now covers most of the United States. If you're planning a trip to San Francisco, for example, you can access OpenTable.com, select the area of the city you'll be visiting, and view descriptions, reviews, customer ratings, and, in most cases, the menus of the restaurants in the area. You can then make a reservation at the restaurant and print a map and the directions to it. The basic tasks that OpenTable.com helps people perform have always been done: looking for a restaurant, comparing prices and menus, soliciting advice from people who are familiar with competing restaurants, and getting directions. What OpenTable.com does is help people perform these tasks in a more convenient and expedient manner.

Another aspect of technological advances is that once a technology is created, products often emerge to advance it. For example, the creation of the Apple iPhone, iPad, and similar devices has in turned spawned entire industries that produce compatible devices. For example, Rokit Boost is a high-end mobile accessories company that makes smartphone cases, headphones, portable USB device chargers, and Bluetooth speakers. Rokit Boost wouldn't exist if it weren't for the advent of the smartphone industry. Similarly, there are a growing number of start-ups working on smartphone apps. An example is Lumbo Body Tech, the subject of the "You Be the VC 10.2" feature. Lumbo Body Tech has made a smartphone app that analyzes a user's running form and offers personalized coaching to help the user exceed his or her running goals.

Political Action and Regulatory Changes. Political and regulatory changes also provide the basis for business ideas. For example, new laws often spur start-ups that are launched to take advantage of their specifications. This happened with the initial passage of the Affordable Care Act (ACA) in the United States in 2010. The combination of new regulations, incentitives for doctors and hospitals to shift to electronic records, and the release of mountains of data held by the Department of Health and Human Services (on topics such as hospital quality and nursing home patient satisfaction), yielded opportunities for entrepreneurs to launch electronic medical records' start-ups, apps to help patients monitor their medications, and similar companies.[10] Changes proposed for the ACA by the new administration that entered the White House on January 20, 2017 also yield opportunities for entrepreneurs to pursue.

On some occasions, entire industries hinge on whether certain government regulations evolve in a manner that is favorable to the industry. For example, there are several start-ups poised to commercialize the use of drones, or Aerial UAV's. Drones can potentially be used for a number of domestic purposes, such as helping farmers determine the optimal level of fertilizer to place on crops, delivering medicine to remote locations, and helping filmmakers shoot overhead shoots. As of June, 2016, the FAA permitted drones to be used only under certain conditions, such as within eyesight of the operator, maximum weight of 55 pounds, daylight-only operations, and a maximum altitude of 400 feet above ground. Several companies, including CyPhyWorks, the subject of the "You Be the VC 8.1"

feature, are working on more advanced applications, such as drones delivering packages in residential neighborhoods. Drone start-ups are waiting for the FAA to develop more liberal rules and standards that will allow drones to be used for expanded purposes. Drone technology will also need to advance before farsighted applications like package delivery are possible and economically attractive.

Political change also engenders new business and product opportunities. For example, global political instability and the threat of terrorism have resulted in many firms becoming more security-conscious. These companies need new products and services to protect their physical assets and intellectual property, as well as to protect their customers and employees. An example of a start-up in this area is Evolv, a Waltham, MA-based startup funded in part by Bill Gates. Launched in 2013, Evolv assembled a multidisciplinary team of experts to identify, invent, and apply new technologies to meet current terrorism-related threats. Combining sensors, artificial intelligence, and human IQ, the company developed Evolv Mosaiq, a security platform which is currently one of the world's most advanced threat detection systems.[11]

Table 2.1 offers additional examples of changes in environmental trends that provided fertile soil for opportunities and subsequent concepts to take advantage of them.

TABLE 2.1 Examples of How Changes in Environmental Trends Provided Openings for New Business and Product Opportunities

Changing Environmental Trend	Resulting New Business, Product, and Service Opportunities	Companies That Resulted
Economic Trends		
Rising personal incomes	Technology products, travel services, organic food	Jawbone, Hipmunk, Whole Food Markets
Baby boomers entering retirement with disposable income	In-home care, health and wellness apps, financial services for older people, travel-related services for older people	Honor, Guardian Medical Monitoring, Betterment, Senior Travel Services
Social Trends		
Increased interest in different, tastier, and healthier food	Healthy-fare restaurants, ethnic packaged foods, craft beer, functional beverages	Froozer, Beyond Meat, Hint (bottled water), Popchips
Increased interest in fitness as the result of new medical information warning of the hazards of being overweight	Fitness centers, pilates and yoga studios, exercise apps, weight-loss programs and apps	SoulCycle, 24 Hour Fitness, Yogaview (Chicago), Peloton, Noom
Technological Advances		
Smartphones	Smartphone operating systems, smartphone apps, smartphone accessories	Android, Instagram, Snapchat, Rokit Boost, iSkin
Wearable technology	Wearable fitness devices, wearable lifesytle devices (like Google Glasses), and wearable tech gadgets	Apple Watch, Spectacles, Oculus Rift, Recon Jet
Political and Regulatory Changes		
Increased EPA and OSHA standards	Consulting companies, software to monitor compliance	MobileEpiphany, SWAC, ESS Group, Air Quality Consultants
Affordable Care Act	Electronic health records, new models of primary care, apps and technologies to deliver care more efficiently	CareCloud, VillageMD, Aledade, Ginger.io, Dispatch

Solving a Problem

The second approach to identifying opportunities is to recognize problems and find ways to solve them. Problems can be recognized by observing the challenges that people encounter in their daily lives and through more simple means, such as intuition, serendipity, or chance. There are many problems that have yet to be solved. Commenting on this issue and how noticing problems can lead to recognizing business ideas, Philip Kotler, a marketing expert, said:

> Look for problems. People complain about it being hard to sleep through the night, get rid of clutter in their homes, find an affordable vacation, trace their family origins, get rid of garden weeds, and so on. As the late John Gardner, founder of Common Cause, observed: "Every problem is a brilliantly disguised opportunity."[12]

Consistent with this observation, many companies have been started by people who have experienced a problem in their own lives, and then realized that the solution to the problem represented a business opportunity. For example, in 1991, while leaving a local coffee drive-thru as he was taking his daughter to school, Jay Sorensen dropped a cup of coffee in his lap because the paper cup was too hot. This experience led Sorensen to invent an insulating cup sleeve and to start a company to sell it. Since launching his venture, the company, Java Jacket, has sold over four billion cup sleeves. Similarly, after watching countless women walk home barefoot after a long night in heels, New York University finance students Katie Shea and Susie Levitt started a company named CitySlips to make easily portable, comfortable shoes. They created a pair of flats that fold up to fit into a pocket-size zip pouch, which easily fits into most women's purses. When a woman pops on the shoes, the pouch unfurls into a tote bag to carry the high heels. The two began selling CitySlips in 2009; today, their product is carried in over 500 stores.[13]

Advances in technology often result in problems for people who can't use the technology in the way it is sold to the masses. For example, some older people find traditional cell phones hard to use: the buttons are small, the text is hard to read, and it's often difficult to hear someone on a cell phone in a noisy room. To solve these problems, GreatCall, Inc. is producing a line of mobile phones targeted to sell to older users. The Jitterbug Flip, for example, is a simple and easy-to-use cell phone while the Jitterbug Smart was the largest smartphone the firm produced in 2017. Both phones feature large keypads that make dialing easy, powerful speakers that deliver clear sound, easy-to-read text, and simple text-messaging capability. Another company that was founded in Australia, Kisa, has created a cell phone designed specifically for kids. The phone has a simple design and allows children to call up to 10 pre-set phone numbers. The back of the phone features an SOS button and can include medical or contact information. The large, bright display can be customized in a number of ways, including to display the names and photos of the people the child is allowed to call. For example, the screen might display boxes that say Mom, Dad, Grandma, Grandpa, Aunt Sarah, Uncle Jason, and so forth. All the child has to do is press the box to make a call. The phone is light enough that it isn't a burden for a child to carry, but durable enough to withstand a fall or two on the playground.[14]

Some business ideas are gleaned by recognizing problems that are associated with emerging trends. For example, sales of traditional low-cal sodas, like Diet Pepsi and Diet Coke, are down, largely due to growing concerns that people have with consuming artificial sweeteners. This trend has left an opening for start-ups such as Sparkling Ice and Spindrift Soda, which make carbonated beverages that are tasty but are made from natural ingredients. Similarly, the proliferation of smartphones enables people to stay better connected, but results in problems when people aren't able to access electricity to recharge their phones for a period of time. A number of companies have solved this problem in innovative ways. An example is the $70 Eton BoostTurbin 2000, which is a device that weighs less than four ounces. It combines a 2000-mAh battery with

a hand crank that provides enough power to fully charge a typical smartphone. The device itself can be recharged via a USB port or the hand crank. The $155 HiNation HiLight is a little bulkier but packs more power and has multiple functions. It has a battery that can recharge an iPhone at least three times. The device itself can be recharged via solar power in about 10 hours.

On some occasions, individuals see problems that are rather obvious, but for some reason, no one has addressed them in a meaningful way. With a little ingenuity, a business can be created. For example, Jeannine Fradelizio liked to entertain. Because she lives close to Northern California's wine country, her parties always include good wine. One thing Ms. Fradelizio noticed is that when people put down their wine glass, they often forgot which one was theirs, so they would abandon the drink or reach for somebody else's. To solve this problem, she initially grabbed a marker from her drawer and scribbled people's names on the wine glasses. People liked the idea and Ms. Fradelizio realized she was on to a business idea. She needed a marker that was nontoxic, odorless, and wouldn't interfere with the bouquet of the wine. In addition, it needed to dry quickly without smudging but also needed to wash off easily. She enlisted the help of a chemist, and about a year later, the Wine Glass Writer was born. That was 2011. Today, Wine Glass Writers are sold at wineries, cooking stores, and other retailers in the United States and several foreign countries. Sales for 2015 topped $1.5 million.[15]

Additional examples of people who launched businesses to solve problems are included in Table 2.2.

Social ventures, a new type of entrepreneurial venture, are launched for the purpose of solving a range of social problems including poverty, hunger, and a lack of access to services such as water and electricity. These ventures can be for-profit firms or not-for-profit agencies. Many businesses are created when an attempt to solve a problem results in an opportunity being identified.

TABLE 2.2 Businesses Created to Solve a Problem

Entrepreneur(s)	Year	Problem	Solution	Name of Business That Resulted
Annie Halsall and Sara Maskopf	2015	In an unfamiliar area, there is no easy way for parents to know the retail locations and restaurants that are stroller friendly.	Create an online "Yelp"-like platform that allows parents to rate locations on "child-friendly" attributes	Winnie (www.winnie.com)
David Zhang and Clement Perrot	2014	No way to instantly have a physical photo to share with friends from a picture taken on a smartphone	Create an instant camera case for smartphones that can print a beautiful photo within seconds	Prynt (www.pryntcases.com)
Parking Panda	2011	There is no easy way to find a parking place in a city, particularly for a concert or a major sporting event	Create a web-based platform that allows users to find and reserve guaranteed parking in a city	Parking Panda (www.parkingpanda.com)
Sherwin Sheik	2011	It's not easy to find reliable caregivers for elderly people who would like to stay in their homes, and most professional caregiver services are expensive	Build an online platform that empowers families to easily find and hire caregivers who match their specific needs and budgets	CareLinx (www.carelinx.com)
Perry Chen, Yancey Strickler, and Charles Adler	2009	No easy-to-access platform for funding creative projects, like indie films, record albums, or food-related projects	Create a web-based "crowdfunding" platform that helps artists, musicians, and people involved in other creative projects raise money from the public	Kickstarter

SAVVY ENTREPRENEURIAL FIRM

Careem: Mobilizing a Solution to a Problem

Careem, the Middle East's first and leading taxi-booking app-based service, was created as a result of two entrepreneurs' attempts to a solve a region-wide problem. Mudassir Sheikha, a 39-year-old Stanford-educated Pakistani, and his Swedish colleague Magnus Olsson travelled extensively in their capacity as consultants for McKinsey & Company. During their travels to the Middle East and North Africa (MENA) region, they struggled to find an easy, quick, efficient, and reliable means of transportation. It also quickly became apparent that although Uber had provided a great solution to commuters' needs in many markets around the world, it had not yet entered the MENA region, and this presented an attractive opportunity. Inspired by the global transformation of taxi-booking services and recognizing an underserved market, the two tech-savvy consultants launched Careem in the city of Dubai, the United Arab Emirates, in 2012. Although Careem was initially launched for business customers, the company soon found that its users were booking rides for personal use, like airport trips, schools drops, and pick-ups.

Expanding Regionally

Careem is touted as one of the UAE's most successful start-ups, with growth of 30 percent month on month. Within five years of its launch, it has expanded to 60 cities across 11 countries, claims over 10 million registered users, and has over 250,000 contract drivers. Careem has ambitious plans to become leading transporter of people and things in the Middle East. With car ownerships as low as 2 to 5 percent in some countries and a large population of 700 million people in the region, there is a potential of between 150 million and 200 million trips a day—a huge opportunity for Careem.

Taking the Home Advantage

As a homegrown player, Careem has unmatched insight into the region's consumers, topography, infrastructure, neighborhoods, roads, and local commuting needs, allowing it to quickly act on the opportunity to develop better maps than and have a competitive advantage over Uber. Many cities and their neighborhoods in the region were inadequately covered by Google Maps and Nokia Maps, so Careem sent its own team to build a location database. The brand name, which means "generous" in Arabic, also worked very well with the local consumers.

Playing the Local Favorite

Close familiarity with the cultural and traditional values helped Careem to serve the regional markets well. In Saudi Arabia, where women were banned from driving until October 2017, Careem had a massive female following, making up nearly 80 percent of its customers. It even introduced a pre-booking facility for women so they could conveniently plan their trips. In Dubai, Careem launched its Ameera services, a ladies-only ride with a woman chauffer. In September 2016, in Saudia Arabia, Careem offered a special service during Eid-al Adha, a popular Muslim festival during which an animal, usually a sheep or a goat, has to be to sacrificed and distributed to family, friends, and the poor. This requires trips to farms, where they are sacrificed and then brought home. Careem offered to provide this service: its drivers got the animal, sacrificed it for the customers, and brought it to customers' homes according to the tradition. In situations where customers wanted the live animal to be transported to their homes, Careem arranged for pick-up vehicles for that too.

Dealing with the Rivalry

Uber entered the Middle East market in 2013 as a global brand with international appeal, changing the competitive dynamics in a growing market that presented an attractive opportunity. Both Uber and Careem adopted competitive strategies to capitalize on that opportunity. Careem already had a strong hold on the market due to its first-mover's advantage and local orientation. Due its global dominance, Uber was able to a raise $15 billion in venture capital compared to $425 million that Careem had raised. Faced with a changing competitive scenario and an aggressive opponent, Careem focused on providing more localized solutions to the region, including better maps, pre-booking services, dedicated call centers, and a varied fleet. Careem claims that other services operate only as a limo service, but it offers a different ride experience for everyone. For example, in Dubai, Careem offered different types of rides: Economy, Business, Careem Kids, MAX, First Class, and Ameera. While Uber seeks global dominance, Careem is concentrating on being a leading regional player with a strong focus on local needs and wants.

Questions for Critical Thinking

1. Identify the market gaps that Shiekha and Olson filled by launching Careem in the Middle East. What role did their personal commuting experience in the region play in identify the opportunity?
2. Comment on how being a local player helped Careem to cater to the needs of users in the region. What factors have contributed to its regional popularity?
3. As the taxi-booking segment matures in the Middle East, how relevant will factors like innovativeness and first-mover's advantage be to sustainability and continued success?
4. Considering the strong competition from global leader Uber, do you think Careem will be able to maintain its hold in the region?

Sources: Frank Kane, "Gentlemen, Start Your Engines: It's Uber vs Careem," *The National*, November 24, 2015, available at https://www.thenational.ae/business/gentlemen-start-your-engines-it-s-uber-vs-careem-1.13430 (Accessed November 7, 2017); Parmy Olson, "Meet the Unicorn That's Beating Uber in The Middle East," Forbes, June 15, 2017, available at https://www.forbes.com/sites/parmyolson/2017/06/15/dial-a-caravan/ (Accessed November 7, 2017).

"The Savvy Entrepreneurial Firm" presents the case of Careem, a car-booking app in the Middle East, which was launched by two ex-McKinsey employees who faced problems finding efficient means of transport in the region.

Finding Gaps in the Marketplace

Gaps in the marketplace are the third source of business opportunities. There are many examples of products that consumers need or want that aren't available in a particular location or aren't available at all. Part of the problem is created by large retailers, such as Walmart and Costco, which compete primarily on price and offer the most popular items targeted toward mainstream consumers. While this approach allows the large retailers to achieve economies of scale, it leaves gaps in the marketplace. This is the reason that clothing boutiques, specialty shops, and e-commerce websites exist. These businesses are willing to carry merchandise that doesn't sell in large enough quantities for Walmart and Costco to carry.

Product gaps in the marketplace represent potentially viable business opportunities. For example, Tish Cirovolo realized that there were no guitars on the market made specifically for women. To fill this gap, she started Daisy Rock guitars, a company, which makes guitars just for women. Daisy Rock guitars are stylish, come in feminine colors, and incorporate design features that accommodate a woman's smaller hand and build. In a related manner, Southpaw Guitars located in Houston, Texas, carries only guitars that are designed and produced for left-handed players. A start-up in a completely different industry is TopTal, the subject of the "You Be the VC 9.1" feature. TopTal fills a gap in the online recruiting industry by supporting a platform that enables companies to find top-notch software engineering developers and designers who are rigorously screened and are willing to work on a contract basis.

Additional examples of companies that started to fill gaps in the marketplace are provided in Table 2.3.

Gaps in the marketplace are commonly recognized when people become frustrated because they can't find a product or service they need and recognize that other people feel the same way. This scenario played out for Lorna Ketler and Barb Wilkins, who became frustrated when they couldn't find stylish

stockbroker/123rf.com

There are many gaps in the marketplace to be filled. This entrepreneur launched a start-up that runs after-school programs for students interested in STEM (Science, Technology, Engineering, and Mathematics) projects. These children recently competed in a national robotics competition for students in their age group.

TABLE 2.3 Businesses Created to Fill a Gap in the Marketplace

Gap in the Marketplace	Resulting New Business Opportunity	Name of Businesses That Resulted
No fitness centers that are open 24 hours a day	24-hour fitness centers to accommodate people who work odd hours	Anytime Fitness, 24 Hour Fitness, Snap Fitness
Lack of toys and toy stores that focus on a child's intellectual development	Toy stores, toy manufacturers, websites that sell educational toys, and toy and smartphone app combinations	Wonder Workshop, Little Bits, Osmos Genius Kit, Ubolly
Too few women pursuing careers in engineering	For-profit and non-profit organizations that teach older girls to code, after-school programs that engage school-age girls in engineering-related projects, interactive books and games that interest young girls in engineering	Girls Who Code, Engineering for Kids, GoldieBlox, Roominate
Shortage of clothing stores that sell fashionable clothing for hard-to-fit people	Boutiques and retail stores that sell fashionable clothing for hard-to-fit people, including plus-sized clothing, maternity clothes, or clothing for tall or short people	Casual Male XL, Fashions to Figure, Motherhood Maternity

"plus-sized" clothing that fit. In response to their frustration, they started Bodacious, a store that sells fun and stylish "plus-size" clothing that fits. Ketler's and Wilkins' experience illustrates how compelling a business idea can be when it strikes just the right chord by filling a gap that deeply resonates with a specific clientele. Reflecting on the success of Bodacious, Wilkins said:

> It's so rewarding when you take a risk and it pays off for you and people are telling you every single day, "I am so glad you are here." We've had people cry in our store. It happens a lot. They're crying because they're so happy (that they're finding clothes that fit). One woman put on a pair of jeans that fit her, and she called me an hour later and said, "They still look good, even at home!" Sometimes people have a body change that happens, whether they have been ill or had a baby, and there's lots of emotion involved in it. If you can go and buy clothes that fit, that helps people feel good about themselves.[16]

Sometimes gaps in the marketplace are noticed via serendipity or chance, similar to the way that problems that need to be solved are spotted. An example is the creation of Simple Gum Natural Chewing Gum. In 2012, Caron Proschan had just finished lunch of green juice and salad with a friend when she reached into her pocket for a stick of gum. When she looked at the gum, she realized it was a stark contrast to the healthy meal she had just eaten. She started looking for a healthy gum, but couldn't find one. Not to be deterred, she decided to create one. She tried to find an appropriate recipe and someone to manufacture a "healthy gum" for her, but struck out on both counts. She struck out on her own and eventually developed a gum that satisfied her. A year later, Simply Gum was born. It is now in approximately 1,200 stores across the country and is consistently a top seller in the chewing-gum category on Amazon.com.[17]

One thing entrepreneurs must remain mindful of in pursuing business opportunities, regardless of whether the opportunity results from changing environmental trends, solving a problem, or finding gaps in the marketplace, is that the opportunity must ultimately be fashioned into a successful business. The nearby "What Went Wrong?" feature focuses on Everpix, a company that resulted from its founders' frustration regarding the lack of a good service to store and organize photos. Regrettably, the founders spent too much time focused on the opportunity at the expense of the business, as you'll see in the feature.